Retail Customer Purchasing Patterns

A data analysis report built with Python, PostgreSQL and Power BI

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1. Purpose of the Study

The study examines a retail dataset of 3,900 individual transactions covering several product categories. Its aim is to understand how much customers spend, which groups they fall into, what they prefer to buy, and how membership subscriptions affect their behaviour, so that the findings can support practical business decisions.

2. About the Data

- **Size:** 3,900 rows and 18 columns.
- **Customer profile fields:** age, gender, location and subscription status.
- **Transaction fields:** item bought, category, amount paid, season, size and colour.
- **Behavioural fields:** whether a discount or promo code was used, number of previous purchases, how often the customer buys, review rating and shipping type.
- **Data quality:** 37 entries were missing from the Review Rating column.

3. Data Preparation and Exploration in Python

The work started by loading and tidying the data in Python. The steps were:

- **Loading:** the file was read into a DataFrame with `pandas`.
- **First look:** `df.info()` was used to review the structure and `.describe()` to produce summary statistics (shown below).
- **Handling gaps:** null values were located, and each missing `Review Rating` entry was filled with the median rating of its product category.
- **Naming convention:** all column headers were converted to snake_case so they are easier to read and document.
- **New fields:** an **age_group** column was derived by grouping customer ages into bands, and a **purchase_frequency_days** column was derived from the purchase-frequency information.
- **Redundancy test:** `discount_applied` and `promo_code_used` were compared, found to duplicate each other, and `promo_code_used` was removed.
- **Moving to SQL:** the cleaned DataFrame was pushed from Python into a PostgreSQL database for querying.

Summary statistics (part 1)

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

Summary statistics (part 2, remaining columns)

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

4. Business Questions Answered with SQL

Ten questions were put to the PostgreSQL database. Each is listed below with the output it returned.

1. Revenue split by gender – total sales were summed separately for female and male customers and compared.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. Big spenders who still used a discount – this query lists customers who redeemed a discount yet paid more than the overall average order value (839 customers matched; the first rows are shown).

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:00.133

3. Five best-rated items – items were ranked by their mean review score.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Standard versus Express delivery – the mean order value was calculated for each of the two shipping options.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers against non-subscribers – customer counts, mean spend and overall revenue were compared between the two groups.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Items most reliant on discounts – the five products with the largest share of orders placed with a discount were identified.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Grouping customers by loyalty – each customer was placed in the New, Returning or Loyal segment according to purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Most ordered items within every category – the three best-selling items were listed for each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat purchasers and subscription – the query checked whether customers with more than five purchases tend to hold a subscription.

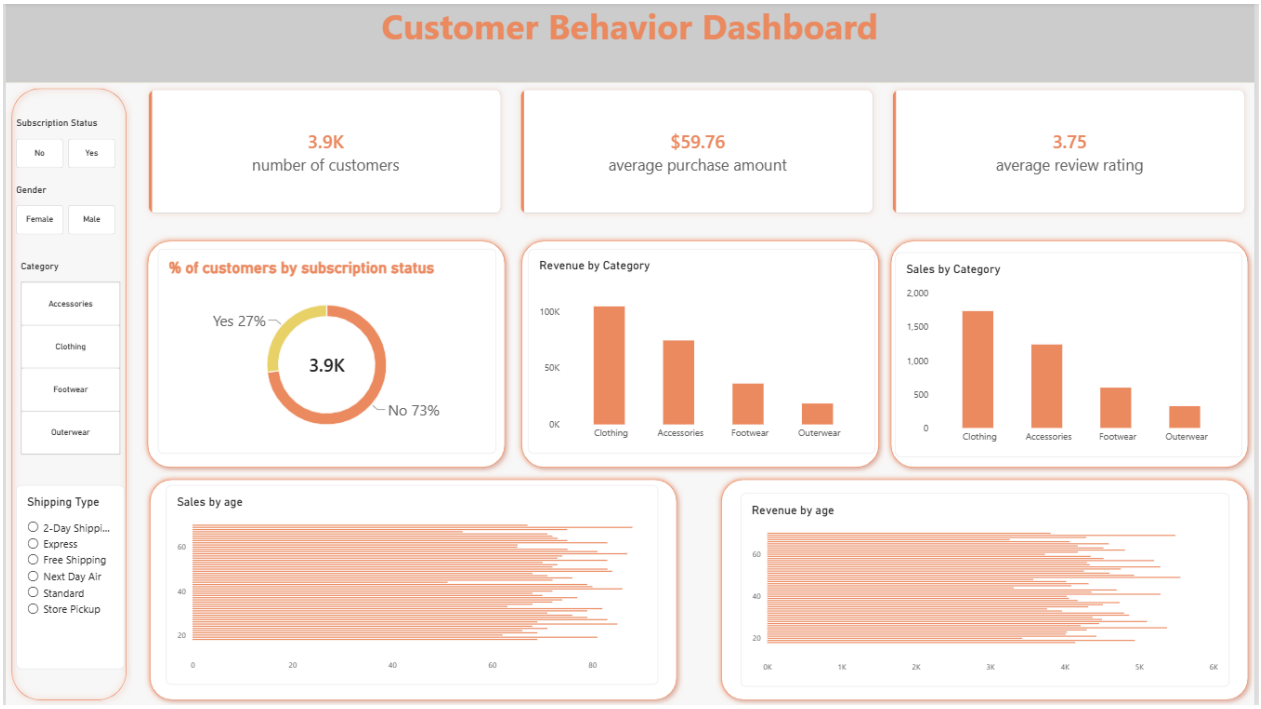
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue contributed by each age band – total revenue was added up for every age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Interactive Power BI Dashboard

To make the results easy to explore, an interactive dashboard was created in Power BI. It pulls the main figures and charts onto one screen and lets viewers filter by subscription status, gender, category and shipping type.



6. Recommendations for the Business

- **Grow the subscriber base** – advertise the special perks that come with subscribing.
- **Build a loyalty scheme** – offer rewards to repeat shoppers so that more of them reach the “Loyal” segment.
- **Revisit discounting** – weigh the extra sales that offers bring in against the effect on profit margins.
- **Feature the winners** – give top-rated and best-selling items prominent space in promotions.
- **Aim campaigns carefully** – concentrate marketing on the age groups that earn the most and on customers who choose express shipping.